Energy Decay in a Damped Harmonic Oscillator

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Abstract

We study the free motion of a mass on a linear spring subject to viscous damping. We derive the closed-form solution in the underdamped regime, show that the mechanical energy decays monotonically, and compare the analytic solution with two simple numerical schemes.

1 Introduction

The damped harmonic oscillator is the simplest model of a system that loses energy to its surroundings [1]. Its equation of motion is

$$m\ddot{x} + c\dot{x} + kx = 0, \tag{1}$$

where $m > 0$ is the mass, $c \geq 0$ the damping coefficient and $k > 0$ the spring constant.[1] Dividing by m and writing $\omega_0 = \sqrt{k/m}$ and $\gamma = c/(2m)$ gives the *standard form*

$$\ddot{x} + 2\gamma\dot{x} + \omega_0^2 x = 0. \tag{2}$$

2 Solution

Substituting $x = e^{\lambda t}$ into (2) gives the characteristic equation $\lambda^2 + 2\gamma\lambda + \omega_0^2 = 0$, whose roots are

$$\lambda_{\pm} = -\gamma \pm \sqrt{\gamma^2 - \omega_0^2}. \tag{3}$$

The behaviour depends on the sign of the discriminant:

$$\text{regime} = \begin{cases} \text{underdamped}, & \gamma < \omega_0, \\ \text{critically damped}, & \gamma = \omega_0, \\ \text{overdamped}, & \gamma > \omega_0. \end{cases}$$

In the underdamped case let $\omega_d = \sqrt{\omega_0^2 - \gamma^2}$. With $x(0) = x_0$ and $\dot{x}(0) = 0$ the solution is

$$x(t) = x_0 e^{-\gamma t}\left(\cos\omega_d t + \frac{\gamma}{\omega_d}\sin\omega_d t\right). \tag{4}$$

Equivalently, with $\mathbf{y} = (x, \dot{x})$, equation (2) becomes the first-order system

$$\dot{\mathbf{y}} = \begin{pmatrix} 0 & 1 \\ -\omega_0^2 & -2\gamma \end{pmatrix}\mathbf{y}. \tag{5}$$

Theorem 1. *The energy $E(t) = \tfrac{1}{2}m\dot{x}^2 + \tfrac{1}{2}kx^2$ never increases. More precisely, $\frac{dE}{dt} = -c\dot{x}^2 \leq 0$.*

[1] All quantities are in SI units.

Proof. Differentiate E and use (1): $\square$

$$\begin{aligned}\frac{dE}{dt} &= m\dot{x}\ddot{x} + kx\dot{x} \\ &= \dot{x}(m\ddot{x} + kx) = -c\dot{x}^2.\end{aligned}$$

3 Numerical check

We integrated (2) with $\omega_0 = 2$, $\gamma = 0.2$ and $x_0 = 1$ on $t \in [0, 10]$, using the explicit Euler method and the classical fourth-order Runge–Kutta method (RK4) [2].

For each step size h we:

1. integrated from $t = 0$ to $t = 10$ with $10/h$ steps;
2. compared every step with the exact solution (4);
3. recorded the largest absolute difference.

Table 1 lists the results and Figure 1 shows the exact solution.

Table 1: Maximum absolute error on $[0, 10]$.

Step h	**Euler**	**RK4**
0.100	8.51×10^{-1}	5.00×10^{-5}
0.050	2.50×10^{-1}	3.10×10^{-6}
0.025	1.06×10^{-1}	1.93×10^{-7}

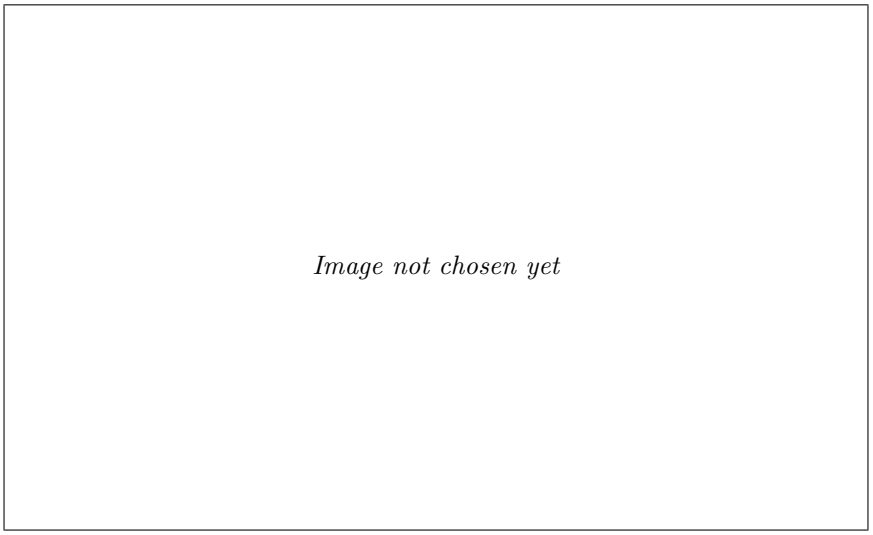


Figure 1: The solution (4) for $\gamma = 0.2$, $\omega_0 = 2$ and its envelope $\pm e^{-\gamma t}$.

Two things stand out:

- Euler's error is large because the method **adds** energy at every step, which works against the physical damping.
- Halving h divides the RK4 error by about $16 = 2^4$, as expected for a fourth-order method.

References

[1] H. Goldstein, C. Poole and J. Safko, *Classical Mechanics*, 3rd ed., Addison-Wesley, 2002.

[2] E. Hairer, S. P. Nørsett and G. Wanner, *Solving Ordinary Differential Equations I*, 2nd ed., Springer, 1993.